How accurately can Hanzi size predict Chinese proficiency: Evidence from learners of Chinese as a second/foreign language

Abstract Diverse measures have been used as indicators of L2 learners' proficiency for research and instructional purposes, yet the accuracy of these measures in predicting L2 proficiency has not been examined. To fill this gap, the present study attempted to investigate the predictive accuracy of Hanzi (Chinese characters) size on HSK (a standardized L2 Chinese proficiency test) level among 403 learners of Chinese as a second/foreign (CS/FL) language. Two major findings emerged. First, the results of Hanzi size test revealed that CS/FL learners who obtained certificates at HSK levels 4, 5, and 6 were estimated to recognize approximately 1,134, 1,637, and 2,271 Hanzi, respectively, with an incremental increase of about 500–600 Hanzi between each level. Second, the overall predictive accuracy of Hanzi size on HSK level was approximately 60%, with higher accuracy for HSK Level 4 than HSK Levels 5 and 6. Based on these findings, implications for future research involving measures of L2 proficiency and Hanzi acquisition were discussed.

Keywords: Hanzi recognition test, Hanzi size, Hanzi instruction, Hanzi acquisition, Chinese as a second/foreign language

Introduction

A critical focus in second language acquisition (SLA) research is measuring L2 learners' proficiency, which has been assessed by diverse methods (Hulstijn, 2012; Park et al., 2022; Thomas, 1994; Zhang, 2018). One commonly used method is the measure of a specific aspect of linguistic component, such as vocabulary size test. In contrast to the widespread use of vocabulary size test in sound-based writing systems, Hanzi (Chinese character) size, defined as the number of Hanzi an individual can recognize, has often been utilized to assess Chinese proficiency in research and placement contexts (Gao, 2017; Jiang & Liu, 2004; Kang & Wu, 2024; Li & Wu, 2024; Song et al., 2024; Wu, 2024; Wu et al., 2017; Zhang, 2018). This distinction mainly stems from the unique relationship between Hanzi and words in Chinese. Chinese words are composed of morphosyllabic Hanzi, each of which

generally corresponds to one syllable and one morpheme. A single Hanzi may function independently as a word or combine with others to form multi-Hanzi words, such as 人 (rǐn, person), 人民(rǐnmín, people) and 人民币 (rǐnmínbì, Chinese Yuan). Consequently, acquiring a critical threshold of Hanzi is regarded as foundational for the development of literacy skills for learners of Chinese as a second/foreign language (CS/FL)[1].

Although researchers have employed Hanzi size test to represent CS/FL learners' global Chinese proficiency, two critical questions remain unresolved. The first question is about Hanzi size of CS/FL learners with different proficiency levels measured by standardized HSK[2] test. The second question concerns the predictive accuracy of Hanzi size on CS/FL learners' HSK level. Existing studies have not systematically explored these questions. Resolving these gaps is critical for elucidating the developmental trajectory of CS/FL learners' Hanzi recognition skills, and advancing empirical research on Hanzi acquisition. Furthermore, the findings could offer further empirical evidence for the validity of using Hanzi size as a measure of L2 Chinese proficiency for research and instructional purposes.

Literature review

The literature review section concentrates on two key topics: measures of L2 proficiency and empirical survey of CS/FL learners' Hanzi size.

Measures of L2 proficiency

Researchers have conducted numerous studies to explore the components of L2 proficiency, with widespread acknowledgement that communicative ability is its core. In exploring the essence of communicative language ability and how second language proficiency enables effective communication across diverse social and

[1] The term *Chinese as a Second/Foreign Language (CS/FL)* was used because some learners may have the experience of learning Chinese both as a second and foreign language, owing to the increasing international exchange. Thus, it is quite challenging to distinguish between learning CSL and CFL solely based on the learners' context.

[2] HSK (Hanyu Shuiping Kaoshi) is a standardized Chinese proficiency test, and comprises nine levels, with Levels 1–6 introduced in 2009 and Levels 7–9 added in 2022 (test for Levels 79 has not been widely administered). Higher levels correspond to greater Chinese proficiency. It was estimated that HSK levels 1–6 approximately align with the CEFR's six proficiency tiers (A1–C2) and the ACTFL Proficiency Guidelines' ratings from Novice High to Superior for receptive competencies. However, the alignment between the new nine-level HSK and CEFR or ACTFL has not been conducted.

academic settings, researchers have proposed a range of theoretical frameworks for L2 proficiency. Several of these models, such as those by Canale and Swain (1980), Bachman and Palmer (1996), Douglas (2000), and Purpura (2004), explicitly detail the components of L2 proficiency and their hypothesized interconnections. Key similar components outlined in these models include, but are not limited to, grammatical, lexical, discourse, sociolinguistic, and pragmatic knowledge. These models aim to establish a broad theoretical groundwork for assessing L2 proficiency, particularly in contexts where language tests are designed and interpreted across varied real-world usage scenarios.

Inspired by these theoretical models, researchers have developed various standardized L2 proficiency tests, such as TOEFL and IELTS for English, and HSK for Chinese. However, due to the challenges in terms of limited time and efforts faced by researchers and participants in the context of conducting research, existing studies have widely used other methods, instead of solely relying on standardized language tests, to measure participants' L2 proficiency. These methods could be divided into two types (Park et al., 2022). The first type is independent, including standardized language test, measures of a specific aspect of linguistic component (e.g., vocabulary or Hanzi) or skill (e.g., reading, speaking). The second type is non-independent, including classroom level or years of instruction, existing proficiency scores, self-ratings and impressionistic judgement. The results of a recent survey (Park et al., 2022) revealed that researchers relied slightly more on non-independent measures (57.22%) than on independent measures (42.78%).

Similar surveys have also been conducted in CS/FL studies. Zhang's study (2018) among two top CS/FL journals published in China (1970s-2018) revealed that the most commonly used method of assessing L2 Chinese proficiency was classroom level or years of instruction, followed by HSK test, reading comprehension test, Hanzi size test, and listening-speaking test. Among these measures of L2 Chinese proficiency, Hanzi size test has been considered as Chinese-specific because of the uniqueness of Hanzi in Chinese. As mentioned in

the introduction section, Hanzi could function as an independent word or combine with other Hanzi to form multiple-syllabic words. That is to say, Hanzi is the foundation of the development of Chinese literacy skills, such as reading and writing. Thus, Hanzi size test has been widely used to assess CS/FL learners' proficiency for research or placement purposes (Gao, 2017; Jiang & Liu, 2004; Kang & Wu, 2024; Li & Wu, 2024; Song et al., 2024; Wu, 2024; Wu et al., 2017; Zhang, 2018).

Debates, however, still exist about using Hanzi size as a proxy for Chinese proficiency. First, some studies indicate that findings based on Hanzi size may lack the robustness of those derived from standardized HSK scores (Zhang, 2018; Zhang et al., 2020). However, the limited evidence came from the same research team with a focus on phonological awareness and phonetic radical awareness. Second, the importance of Hanzi size for achieving success in HSK tests may vary across proficiency levels. At higher levels of HSK test, advanced literacy skills, such as expanded vocabulary and passage-level reading and writing abilities, become increasingly essential, potentially diminishing the relative importance of Hanzi size. Empirical evidence partially supports this assumption. Wu et al. (2017) observed that the correlation between Hanzi size and HSK scores was statistically significant only among beginner and intermediate learners, with stronger associations at lower proficiency levels. However, existing research has not systematically mapped Hanzi size to specific HSK levels, leaving the predictive accuracy of Hanzi size for determining HSK levels unresolved.

Empirical survey of CS/FL learners' Hanzi size

Prior to examining the results of existing surveys of CS/FL learners' Hanzi size, it is necessary to first review the recommended number of Hanzi to be recognized in Chinese language guidelines. It is worth noting that although the ACTFL Proficiency Guidelines (United States) and the CEFR (Europe) focus on describing what learners are expected to accomplish across different aspects in Chinese and other foreign languages, researchers have attempted to quantify the linguistic items to achieve specific levels in these guidelines due to the

acknowledged importance of quantifying linguistic items for language learning. For instance, some researchers have explored the vocabulary size necessary to achieve CEFR benchmarks in English (Laufer, 2020; Milton & Alexiou, 2009, 2020; Nation, 2006, 2024). As for Chinese, Bellassen (2011) posited Hanzi volume aligned with CEFR levels, proposing approximately 250 for A1, 500 for A2, 800 for B1, 1,500 for B2, 2,200 for C1, and 3,000 for C2.

Table 1 *Summary of recommended number of Hanzi in existing syllabuses/standards*

1992 Syllabus		2009 HSK Syllabus		2010 Syllabus		2021 Standards	
Level	Hanzi volume	Level	Hanzi volume	Level	Hanzi volume	Level	Hanzi volume
1	800	4	1083	1	900	Beginner level	900
2	1604	5	1710	2	1800	Intermediate level	1800
3	2205	6	2635	3	3000	Advanced level	3000
4	2905						

Note. The number of Hanzi for a higher level is cumulative, including those for lower levels.

1992 Syllabus: Graded Vocabulary and Hanzi Syllabuses for Chinese Proficiency Levels (Hanyu shuiping cihui yu hanzi dengji dagang);

2010 Syllabus: Graded Syllables, Characters, and Words for the International Teaching of Chinese (Hanyu guoji jiaoyu yong yinjie hanzi cihui dengji huafen);

2021 Standards: International Chinese Language Education Standards for Chinese Proficiency Levels (Guoji zhongwen jiaoyu zhongwen shuiping dengji biaozhun)

Different from ACTFL and CEFR, Chinese language guidelines in China have outlined target quantities for linguistic items, including syllables, vocabulary, Hanzi, and grammar. However, discrepancies persist across these guidelines regarding the recommended number of Hanzi to be recognized by CS/FL learners

(Table 1) (Liang, 2022). The 1992 Syllabus established an early benchmark by categorizing 2,905 Hanzi into four levels. The later 2010 syllabus expanded to 3,000 Hanzi, redistributed into three levels. The most recent 2021 Standards retained the total of 3,000 Hanzi, and further introduced a three-level structure subdivided into nine bands: with 900 Hanzi for the elementary level (bands 1–3), another 900 for the intermediate level (bands 4–6), and 1,200 for the advanced level (bands 7–9). In contrast, the 2009 syllabus accompanying the new HSK test only specifies the recommended number of words but omits explicit Hanzi recognition requirements. Our analysis of the HSK vocabulary syllabus reveals that levels 4–6 include 1,083, 1,710, and 2,635 Hanzi respectively. It could be seen that researchers have not reached a consensus about the number of Hanzi required for each proficiency level, and empirical studies of CS/FL learners' actual Hanzi size are needed.

Empirical research on CS/FL learners' Hanzi size remains limited and has yielded different results. Some studies focused on learners of Chinese as a foreign language outside China. Zhang (2017) sampled 108 Hanzi from the 2010 Syllabus to investigate Hanzi size among 40 English CS/FL learners in the U.K. and 43 Arabic CS/FL learners in Egypt. Results revealed that, pre-intermediate learners with one year of university-level Chinese learning had an average Hanzi size of approximately 480 (British group) and 630 (Egyptian group). Intermediate learners with two years of Chinese learning at university exhibited higher means: 810 Hanzi for the British group and 1,050 for the Egyptian group. A recent study by Hou (2024) was conducted among 32 CS/FL learners from two universities in the United States. Hou sampled 135 Hanzi from the Hanzi list in the 2021 Standards and reported an average Hanzi size of approximately 380 for learners with one year of Chinese learning.

Some research was carried out among CS/FL learners in China. Li (2003) conducted one of the earliest investigations in this area, examining 42 intermediate and advanced CS/FL learners enrolled at a university. Learners' proficiency levels were determined by institutional status. Li randomly selected 60 Hanzi from the

1992 Syllabus and assessed Hanzi recognition across pronunciation, semantic meaning, and orthographic form. Results indicated an average Hanzi size of approximately 1,000 for intermediate learners and 1,616 for advanced learners. To address the limitation of unvalidated Hanzi size test in Li's study, Zhang et al. (2021) developed a psychometrically robust Hanzi size test based on the 2010 Syllabus among 318 CS/FL learners recruited from different universities. It was found that learners passing HSK Levels 4, 5, and 6 recognized approximately 1,020, 1,470, and 1,980 Hanzi, respectively. However, Zhang et al. did not control the data quality, such as excluding participants who may not have engaged seriously with the informal and low-stakes assessment, which may risk underestimating CS/FL learners' true Hanzi size.

To sum, while the existing research suggest that CS/FL learners' Hanzi size increases with their growing Chinese proficiency, inconsistent findings may be attributed to heterogeneity in CS/FL learners' learning contexts (China vs. outside China), methods of assessing proficiency (institutional status versus standardized HSK evaluations), and measurement tools for Hanzi size (unvalidated versus psychometrically validated instruments). Such methodological variability precludes direct cross-study comparisons, thereby limiting generalizable conclusions about the relationship between CS/FL learners' Hanzi size and their Chinese proficiency (e.g., HSK levels).

The current study

As reviewed above, the key limitation of prior research on the measures of L2 proficiency in SLA area is that the predictive accuracy of measures of a specific linguistic component on standardized L2 proficiency remains unexamined. Similar limitation exists in the measures of L2 Chinese proficiency, and it is not clear how Hanzi size predict CS/FL learners' HSK level. In addition, the Hanzi size of CS/FL learners passing different levels of HSK test still needs more empirical evidence. To address these gaps, this study leverages the standardized HSK test as measures of Chinese proficiency to investigate the following research questions:

RQ1: What are the estimated Hanzi sizes for CS/FL learners passing different

levels of HSK test?

RQ 2: What is the predictive accuracy of CS/FL learners' Hanzi size on their HSK levels?

Method

Participants

This study utilized data from a large-scale Hanzi recognition dataset compiled by the authors, initially comprising 800 CS/FL learners across mainland China. The participants were recruited based on both convenience sampling and snowballing methods, yet we have attempted to balance some essential variables, including L1 background, HSK level, and institutions. To address potential variability stemming from heterogeneous proficiency measures (Zhang, 2018), participants' Chinese proficiency was operationalized as the highest HSK level attained within the two-year period preceding data collection. After applying selection criteria, a final sample of 403 CS/FL learners from 25 universities in 15 provinces was retained (Table 2). The inclusion criteria included both HSK level and Hanzi size restriction. First, participants were required to have passed HSK Levels 4, 5, or 6 within the prior two years. This is because HSK Level 4 has been established as the minimum proficiency benchmark for applying undergraduate programs in China and that HSK Levels 7–9 tests have not been widely administered since the end of 2022.

Table 2 *Demographic information of the participants*

Variable		L1 background	
		Non-Hanzi-culture	Hanzi-culture
Number of participants		246	157
Gender	Male	152	36
	Female	91	121
Age (SD)		22.2(3.31)	21.9(2.79)
Years of Chines learning		3.21(2.95)	4.07(3.89)
Academic degree	Non-degree	27	8
	Undergraduate	182	128
	Master	29	14
Subject background	Chinese-related	179	34
	Others	131	17

HSK level	4	134	48
	5	82	54
	6	30	55

Note. 1. Some participants did not provide complete demographic information.
2. Non-degree programs included Chinese language learning programs, and preparatory programs instructed in Chinese.
3. Chinese-related subjects included those such as Chinese Language and Literature, International Chinese Education, Business Chinese, Traditional Chinese Medicine, as well as exchange and preparatory programs instructed in Chinese.

Second, participants' estimated Hanzi size was required to exceed 60% of the total Hanzi prescribed for their attained HSK level. Specifically, the estimated Hanzi size of learners who passed HSK Levels 4, 5, 6 needed to exceed 650 (1,083 ×.6), 1,026 (1,710×.6) and 1,581 (2,635×.6), respectively. The 60% threshold aligns with the commonly used cut-off score for passing HSK test, where 180 out of 300 marks is considered a passing score for HSK Levels 4 to 6. Applying this cutoff may mitigate underestimation risks inherent to the voluntary, low-stakes administration of the Hanzi size test.

Hanzi size test

This study utilized the Hanzi size test for CS/FL learners developed by Zhang et al. (2021). The test consists of 100 items, sampled using a stratified sampling method from the 3,000 Hanzi in the 2010 Syllabus. These items were categorized into three levels:30 elementary Hanzi, 30 intermediate Hanzi, and 40 advanced Hanzi. The selection of test items was carefully controlled to ensure appropriate difficulty levels and to account for key Hanzi-level characteristics, such as orthographic structure and phonetic regularity. This test has been validated using both Classical Test Theory and Item Response Theory and adopted in existing research (Kang & Wu, 2024; Li & Wu, 2024; Song et al., 2024; Wu, 2024), further demonstrating its validity and applicability.

This test required participants to complete two tasks. The first was the phonological task and involved writing the Pinyin (the phonetic transcription of

Chinese) for the target Hanzi. The second was the semantic task and required the participants to form a word using the target Hanzi. These tasks were designed to comprehensively assess both the phonological and semantic knowledge of the target Hanzi.

As for the scoring criteria, the accuracy was determined based on both phonological and semantic tasks. In the phonological task, only syllable accuracy was assessed, and tonal errors were not considered in the scoring due to the difficulty of tonal acquisition. In addition, excluding tones could lead to stronger validity evidence for assessing Hanzi recognition (Tseng et al., 2016). In the semantic task, responses were rated holistically, emphasizing the overall correctness of the answer. Errors unrelated to the task, such as incorrectly written non-target Hanzi or substituting Pinyin for Hanzi, were disregarded. A score of 1 was awarded only when both phonological and semantic answers were correct; otherwise, a score of 0 was given.

To ensure accuracy and consistency in scoring, two experts participated in the evaluation. First, a random sample of 30 responses was independently scored by both experts based on predefined criteria. The inter-rater reliability was .98, demonstrating a high level of agreement and consistency in their interpretations and the application of scoring standards. Next, the remaining responses were randomly divided into two groups, with each expert independently scoring one group. For any ambiguous responses, the experts resolved discrepancies collaboratively through discussion to maintain scoring reliability. The accuracy of each participant's responses was calculated using the formula: *number of correct answers/100*. To estimate Hanzi size, the accuracy score was then multiplied by 3,000.

Administration

The data collection was conducted in an offline group format, which enabled the administrator to provide timely feedback to the test-takers. During the test, participants were not allowed to consult any reference materials to ensure the integrity of their responses. The total time allocated for completing the test was

limited to 40 minutes.

Results

RQ1: Estimated Hanzi size of CSL learners

As shown in Table 3, the average estimated Hanzi size for the participants who passed HSK Levels 4, 5, and 6 was approximately 1,134, 1,637, and 2,271 Hanzi, respectively, with an average incremental increase of about 500–600 Hanzi between adjacent levels. Based on the 95% confidence interval, the ranges of Hanzi sizes for learners passing HSK Levels 4, 5, and 6 were 1,067–1,201, 1,557–1,718, and 2,181–2,361, respectively. When considering only the number of Hanzi, learners passing HSK Levels 4, 5, and 6 were estimated to have mastered approximately 105% (1134/1083)[3], 95.73% (1637/1710), and 86.19% (2271/2635) of the Hanzi required for each level. Further analysis revealed slight variations between learners from different L1 backgrounds. The average Hanzi size for those who passed HSK Levels 4, 5, and 6 was approximately 1,171, 1,623, and 2,241 for learners from non-Hanzi-cultural backgrounds, and 1,031, 1,658, and 2,288 for those from Hanzi-cultural backgrounds.

Table 3 *Summary of the participants' estimated Hanzi size*

L1 background	HSK level	Mean	95% Confidence Interval	SD
Pooled	4	1,134	1,067–1,201	460
	5	1,637	1,557–1,718	480
	6	2,271	2,181–2,361	425
Non-Hanzi-culture group	4	1,171	1,086–1,256	497
	5	1,623	1,517–1,730	484
	6	2,241	2,072–2,410	451
Hanzi-culture group	4	1,031	938–1,123	318
	5	1,658	1,528–1,788	477
	6	2,288	2,176–2,400	414

RQ2: Predictive accuracy of Hanzi size on HSK level

To address RQ2, we first conducted a two-way ANOVA test, where L1 background and HSK levels were entered as independent variables, and Hanzi size

[3] The first number corresponds to the estimated Hanzi size, while the second reflects the total number of Hanzi prescribed in the HSK vocabulary syllabus at each respective proficiency level.

as dependent variable. The results revealed that the main effect of HSK level on Hanzi size was significant and the effect size was big, $F(2,397)=164.09$, $p<.001$, Partial $\square^2=.45$. However, the main effect of L1 background on Hanzi size was insignificant and the effect size was small, $F(1,397)=.15$, $p=.70$, Partial $\square^2<.001$. In addition, the interaction effect between L1 background and HSK level on Hanzi size was not significant and the effect size was small, $F(2,397)=1.69$, $p=.20$, Partial $\square^2=.008$. These results indicate that it is reasonable to carry out regression analysis based on the pooled data from CS/FL learners in the Hanzi- and non-Hanzi groups.

Following the ANOVA test, a series of ordinal multinomial logistic regression analysis were conducted. In the first model, the dependent variable only included the participants' HSK level (4, 5, or 6, with Level 6 as the reference level), and the independent variable was the participants' Hanzi size. The results of likelihood ratio test indicated that the overall model fit was good, $\square I = 238.23$, $p < .001$. The results showed that Hanzi size was a significant predictor of HSK level (Table 4, Model 1). In addition, the model demonstrated an overall predictive accuracy of 59.31% (Table 5, Model 1), and the prediction accuracies for individual HSK levels was 79.12% for Level 4, 35.29% for Level 5, and 55.29% for Level 6.

Table 4 *Summary of ordinal multinomial logistic regression analysis (Dependent variable=HSK level)*

Model	Variable		Regression coefficient	SE	*z*	Wald □^{2}	*p*	OR
1	Thresholds of the dependent variable	HSK level 4	3.96	.34	11.56	133.97	<.001	.02
		HSK level 5	6.49	.46	14.12	199.31	<.001	.002
	Independent variable	Hanzi size	8.68	.68	12.78	163.34	<.001	5904.49
2	Thresholds of the dependent variable	HSK level 4	6.02	1.24	4.86	23.58	<.001	.002
		HSK level 5	9.13	1.33	6.89	47.52	<.001	<.001
	Independent variables	Age	–.01	.05	–.17	.03	.87	.99
		Gender	–.04	.29	–.14	.02	.89	.96
		L1 background	.97	.26	3.76	14.16	<.001	2.64
		Academic degree	.87	.36	2.38	5.66	.02	2.38
		Subject background	1.03	.42	2.44	5.95	.02	2.85
		Years of CS/FL learning	.29	.07	4.17	17.37	<.001	1.34
		Hanzi size	7.57	.86	8.78	77.13	<.001	1935.78

Note. L1 background: Hanzi-culture group (Code=1), Non-Hanzi-culture group (Code=0); Gender: Male (Code=1), female (Code=0); Academic degree: preparatory program (Code=0), undergraduate (Code=1), master (Code=2); Subject: Chinese (Code=1), others (Code=0). OR=Odds ratio

In the second model, the dependent variable remained the HSK level (4, 5, or 6), while the independent variables included age, gender, L1 background, academic degree, subject background, years of Chinese learning, and Hanzi size. Due to incomplete demographic data from some participants, the final analysis included a total of 335 participants. The results of likelihood ratio test for the second model indicated a good overall fit, □I = 272.90, $p < .001$. The results (Table 4, Model 2) revealed that L1 background, academic degree, subject background, years of CS/FL learning, and Hanzi size were all significant predictors of the HSK level. In addition, the second model demonstrated an improvement over the first model, with an overall predictive accuracy of 63.88%, and the prediction accuracies for individual HSK levels were 74.24% for Level 4, 54.92% for Level 5, and 60.49% for Level 6 (Table 5, model 2).

Table 5 *Summary of predictive accuracy of the regression models*

Model	HSK level	Observed frequency	Predicted accurate frequency	Predictive accuracy
1	4	182	144	79.12%
	5	136	48	35.29%
	6	85	47	55.29%
	In total	403	239	59.31%
2	4	132	98	74.24%
	5	122	67	54.92%
	6	81	49	60.49%
	In total	335	214	63.88%

Discussion

The present study investigated the predictive accuracy of CS/FL learners' Hanzi size on their HSK level among 403 CS/FL learners and revealed two key findings. First, the average Hanzi size for CS/FL learners who obtained HSK Levels 4, 5, and 6 certificates was approximately 1,134, 1,637, and 2,271, respectively. Second,

Hanzi size significantly predicted CS/FL learners' HSK level, yet the predictive accuracy of Hanzi size was higher for HSK Level 4 compared to Levels 5 and 6.

RQ1: Estimated Hanzi size of CS/FL learners

The study revealed that the mean Hanzi size for CS/FL learners attaining HSK Levels 4, 5, and 6 slightly exceeded those reported in prior research (Zhang et al., 2021). This divergence may stem from methodological differences, notably the exclusion of participants whose self-reported HSK levels were incongruent with their estimated Hanzi size. While this selection strategy enhanced internal validity, two limitations warrant acknowledgment. First, the reliance on HSK certification might introduce potential construct-irrelevant variance. The prevalence of multiple-choice formats in HSK assessments may inflate scores through guessing effects, thereby misrepresenting learners' true linguistic competence. In addition, HSK test did not assess CS/FL learners' speaking competence, thus HSK test cannot comprehensively represent CS/FL learners' Chinese proficiency. Second, the dynamic nature of L2 proficiency, which is subject to improvement or attrition following sitting HSK test, could further contribute to the discrepancy between observed Hanzi size and CS/FL learners' Chinese proficiency at the time of data collection. Nevertheless, this approach offers a methodological attempt for estimating Hanzi size while mitigating biases inherent to self-reported proficiency.

The interpretation of the results may vary depending on the reference guidelines or frameworks employed. When analyzed against the HSK syllabus, the results indicate that CS/FL learners at HSK Levels 4, 5, and 6 demonstrate robust mastery of Hanzi prescribed for their respective HSK levels, with proportional mastery relative to HSK syllabi ranging from 86% at Level 6 to 105% at Level 4. However, divergent findings emerge when comparing results with alternative reference frameworks. First, when contextualized within Bellassen's (2011) CEFR-aligned benchmarks, the estimated Hanzi sizes in this study fall short of recommended thresholds. Second, relative to the 2021 Standards, learners at HSK Levels 4 and 5 exhibit Hanzi size corresponding to Bands 4–6 of the intermediate level, while those at Level 6 barely meet the lower band of the advanced level,

falling short of the recommended 3,000–Hanzi benchmark. Third, comparisons with native Chinese-speaking children (Wen et al., 2015) reveal that CS/FL learners at HSK Levels 4–6 demonstrate Hanzi size comparable with elementary students at Grade 2 (n = 1,249) and Grade 3 (n = 2,108) in China, yet below that of Grade 4 students (n = 2,661).

These comparisons might underscore deficiencies in CS/FL learners' Hanzi size to some extent, which may stem from two interrelated factors. First, the objectives of the new HSK framework emphasize reducing assessment rigor and mitigating learner anxiety to broaden its global appeal (Zhang et al., 2010). This divergence from earlier benchmarks has resulted in a contraction of Hanzi requirements: the current HSK syllabus prescribe fewer Hanzi than the 1992, 2010, and 2021 guidelines, potentially exacerbating observed insufficient Hanzi size. Another factor might relate to the backwash effect of HSK test. Different from the 2010 Syllabus and 2021 Standards, which function primarily as reference framework rather than mandatory guidelines for textbook development and classroom instruction, the HSK syllabus tends to exert a greater influence on CS/FL learners, given the importance of obtaining HSK certification for applying to academic programs or for job opportunities. A further analysis of the texts of the sample HSK tests[4] provided by Chinese Testing International reveals constrained Hanzi inventories (Level 4: 922; Level 5: 991; Level 6: 1,412). These results further suggest the potential influence of HSK test on CS/FL learners' Hanzi acquisition.

Although it is not a primary research question of the present study, we would like to briefly discuss the non-significant prediction of L1 background on Hanzi size. This result aligns with previous research, suggesting that for intermediate and advanced CS/FL learners, L1 background may not play a crucial role in Hanzi recognition (Chai & Bao, 2023; Zhang et al., 2021). As CS/FL learners' Chinese proficiency improves, their Hanzi recognition performance tends to become

[4] Retrieved from https://www.chinesetest.cn/HSK on April 4, 2025.

similar to those of native Chinese speakers (Hao, 2018). Similar results have been observed among learners of English as a L2, where the negligible impact of L1 background has been found on word recognition (Lemhцfer et al., 2008). Taken together, these findings suggest that for intermediate and advanced L2 learners, their language acquisition performance is likely to be predominantly influenced by the intrinsic features of the target language, and the role of their L1 background appears to diminish as their proficiency develops. However, this tentative account needs more empirical evidence from diverse L2 learners.

The insignificant prediction of L1 background on Hanzi size among learners from Hanzi-culture and non-Hanzi-culture groups could be attributed to several factors. First, Chinese Hanzi, Japanese Kanji and Korean Hanja share both similarities and differences in terms of form, pronunciation and meaning. However, the orthographic, phonological and semantic differences among the shared Hanzi across the three languages may lead to errors in Hanzi learning (Chen, 2001; Xiao, 2002; Zhang & Xing, 2023). Such discrepancies could potentially offset the advantages of prior exposure to Hanzi for learners from Hanzi-culture regions. Second, the participants might have developed orthographic awareness, which could facilitate the acquisition of intermediate and advanced-level Hanzi (Feng, 2006; Hao, 2007; Jiang, 2001; Lu, 2002; Shen, 2013; Xu, 2007; Zhang, 2007). This claim gains certain support from Zhang & Roberts' (2022) proposition that mastering 1,000 Hanzi constitutes a threshold for the development of phonetic radical awareness among CSL/FL learners. Notably, learners who achieved HSK Level 4 in this study already surpassed this benchmark, suggesting their potential ability to employ orthographic strategies (e.g., phonetic radical analysis) to enhance Hanzi acquisition efficiency. Such metalinguistic competence may mitigate disparities in Hanzi mastery between learners from Hanzi- and non-Hanzi backgrounds, thereby contributing to the observed lack of significant differences in Hanzi size across these groups. However, more empirical evidence is needed in future.

RQ2: The prediction of Hanzi size on HSK levels

This study revealed that CS/FL learners' performance in Hanzi recognition significantly predicted their HSK levels. The regression model that included only Hanzi size achieved an overall accuracy of 59.31% in predicting HSK levels, exceeding the random chance level of 50%. This finding, to some extent, supports the validity of using Hanzi size as an indicator of CS/FL learners' Chinese proficiency (Gao, 2017; Jiang & Liu, 2004; Kang & Wu, 2024; Li & Wu, 2024; Song et al., 2024; Wu, 2024; Wu et al., 2017; Zhang, 2018). Nevertheless, the present study also observed certain limitations in the accuracy of Hanzi size as a predictor of CS/FL learners' HSK levels.

Firstly, the overall predictive accuracy for HSK levels increased from 59.31% to 63.88% after adding additional variables into the model that initially included only Hanzi size. This highlights the potential limitations of relying solely on Hanzi size as an indicator of Chinese proficiency. The primary reason is that Chinese proficiency is a multifaceted construct, encompassing various components, as well as multiple skills. Thus, Hanzi size, while essential, represents only one aspect of this broader proficiency.

Secondly, the results of the two regression analysis models revealed that Hanzi size yielded higher predictive accuracy for HSK Level 4 compared to HSK Levels 5 and 6. This may be attributed to differences in test content across the HSK levels (Peng et al., 2020). HSK Level 4 is considered intermediate, with test content primarily focusing on highly frequent Hanzi and words, basic grammar items, and fundamental sentence writing, all of which are closely tied to basic Hanzi recognition. Notably, after adding other background variables, the predictive accuracy of the regression model for Level 4 decreased from 79.12% to 74.24%, further underscoring the importance of Hanzi size for this level. In contrast, the tests for Levels 5 and 6 assess advanced proficiency, requiring more complex skills, such as long-passage reading and writing. Compared to Level 4, these advanced levels demand a larger vocabulary, stronger mastery of complex grammatical structures, and receptive and productive skills at passage level. This likely explains

the increased predictive power of the second regression model for HSK Levels 5 and 6 after adding additional variables.

Implications

The results of present study could offer significant insights for SLA research. First, the present study might be the first to examine the predictive accuracy of a measure of a specific linguistic component on a standardized Chinese test. This study might present an attempt to examine the relationships among different measures of L2 proficiency. Future research is advised to concentrate on both independent and non-independent measures to depict a clearer picture of the robustness of different measures in representing L2 proficiency.

Second, the findings partially validates the use of Hanzi size as a rapid and effective indicator of Chinese proficiency (Zhang, 2018). Prior studies that evaluated Chinese proficiency using Hanzi size generally divided participants into high- and low-proficiency groups due to the lack of clear alignment between HSK levels and Hanzi size. The present study's findings may provide researchers with a more accurate reference framework, enhancing the precision of inferring Chinese proficiency based on Hanzi size.

Third, the study revealed level-specific differences in the predictive accuracy of Hanzi size on HSK levels. This suggests that Hanzi size may serve as a more reliable assessment indicator for intermediate learners, while presenting certain limitations when evaluating advanced learners' proficiency. This finding has broader implications for the measurement of L2 proficiency in research contexts (Hulstijn, 2012; Park et al., 2022; Thomas, 1994; Tremblay, 2011). It has been observed that differing measures can lead to varying research outcomes (Xu et al., 2022; Zhang, 2018, 2021; Zhang et al., 2020). Given the critical role of accurately measuring L2 proficiency in conducting SLA research, addressing these disparities is essential for improving the comparability, generalizability and replicability of research findings.

The present study may also offer several important implications for Hanzi instruction and research. First, the estimated Hanzi size of CS/FL learners and the

observed non-significant prediction of L1 background on Hanzi size could provide empirical evidence for HSK preparation and Chinese instruction. For CS/FL learners preparing for HSK Levels 4–6, Hanzi size could serve as a reference for assessing their readiness to pass the corresponding levels. In addition, CS/FL learners' Hanzi size might increase by approximately 500–600 between adjacent levels within HSK levels 4-6, and this could help CS/FL teachers manage instructional difficulty effectively by structuring the introduction of new Hanzi in a balanced manner. Furthermore, Hanzi has been identified as a key predictor of Chinese text readability (Du et al., 2022; Jiang et al., 2020; Wu et al., 2020; Yin, 2022; Zuo & Zhu, 2014). Thus, the findings of this study may guide the selection of Hanzi in graded readers, optimizing their readability and facilitating learners' Hanzi intake. The negligible influence of L1 background on Hanzi size might help reduce negative emotions associated with Hanzi learning among CS/FL learners from non-Hanzi-culture regions (Li & Zhang, 2021), and boost their confidence and other positive emotions (e.g., accomplishment and enjoyment). Consequently, Hanzi teaching strategies might focus on the inherent characteristics of Hanzi, instead of learners' L1 backgrounds, particularly for intermediate and advanced CS/FL learners.

Second, the results of the present study suggest the insufficient Hanzi size of CS/FL learners compared with Chinese language guidelines and Chinese-speaking children's performance, and this results has two important implications. The first one is that the insufficient Hanzi size of CS/FL learners may hinder their ability to meet the academic demands required for their studies in Chinese-medium programs. With the implementation of several regulations targeting international students' education in China (Ministry of Education of China, 2020), Chinese universities have increasingly emphasized the quality of international students' enrollment benchmark and graduation theses, which suggests a growing demand for advanced Chinese proficiency in academic context, which requires larger Hanzi size (Ji, 2024; Li et al., 2020; Tao & Chen, 2019; Wang & Jiang, 2019; Zhang, 2024). Another implication relates to the necessity of developing HSK 3.0 (Hankao guoji,

2021). HSK 3.0 is advised to increase the Hanzi volume within its tests to meet the evolving needs of international Chinese education. Such adjustments could enhance the backwash effect of HSK, and further improve CS/FL learners' Hanzi acquisition as well as general literacy skills.

Third, this study may offer empirical evidence to clarify the alignment between Hanzi volume in HSK and proficiency descriptions outlined in frameworks such as the CEFR and ACTFL, alongside other Chinese language tests (e.g., AP Chinese and Stamp). While the CEFR and ACTFL rely on can-do statements, the findings of this research may establish a potential bridge between these qualitative descriptors and quantitative Hanzi requirements. Additionally, these findings may assist educators and researchers in estimating the Hanzi proficiency of CS/FL learners based on non-HSK assessments. For instance, learners achieving C2 in CEFR-based tests or Distinguished proficiency in ACTFL-based tests, both of which are aligned with HSK Level 6, are likely to approximately recognize 2,200 Hanzi. Similarly, AP Chinese, which exhibits intermediate difficulty between HSK Levels 4 and 5 (Lou & Zhu, 2006), might correspond to a range of 1,000–1,600 Hanzi for certified test-takers. To sum, the estimated Hanzi thresholds derived from this study may refine cross-regional alignment among diverse Chinese proficiency frameworks.

Limitations and conclusions

We acknowledge that the present study has several limitations. First, the participants were CS/FL learners who had passed HSK Levels 4–6, excluding learners at HSK Levels 1–3 and 7–9. Furthermore, this study did not incorporate participants' oral proficiency scores from HSKK (Hanyu Shuiping Kouyu Kaoshi, Chinese oral proficiency test), because HSKK and HSK are distinct evaluation systems and HSKK certification was not a prerequisite for academic enrollment or program eligibility in most contexts in China at the time of data collection. As a result, the findings may not fully represent learners across the entire spectrum of global Chinese proficiency. Future research that includes CS/FL learners with a broader range of proficiency could provide a more comprehensive understanding

of this topic. Second, due to the time lag in data collection, there may be some discrepancies between the HSK levels reported by participants and their actual Chinese proficiency at the time of completing the Hanzi size test. This limitation suggests that the findings of this study should be further validated with additional empirical evidence.

Despite these limitations, this study may be among the first to conduct a relatively large-scale and in-depth investigation into the relationship between CS/FL learners' Hanzi size and their HSK level. The overall findings hold significant implications for measuring L2 proficiency for research and instruction purposes, and for Hanzi instruction in the field of international Chinese education. Moreover, the present study provides valuable insights for the further alignment of different guidelines and syllabuses related to Chinese language learning.

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